Automated Optical Inspection In SMT Environment

Automated optical inspection (AOI) is one of the best inspection tools in the SMT environment. It helps identify issues and defects in the early stages of the manufacturing process. This article discusses AOI and its applications in the PCB industry

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The rapid development in the semiconductor industry towards miniaturisation has promoted an apparent change in the printed circuit board (PCB) technology. As a result, the number of pin counts has increased in integrated circuits per unit area. The density of components, track width, and spacing between bonding pads are becoming smaller and smaller. Component pad design, selection of solder paste, and solder volume per pad for good solder joint have become a vital part of electronic PCB assembly production lines.

Further, due to the tremendous increase in production volumes and quality expectations, the process of inspection has become an essential part of the surface-mount technology (SMT) environment. The industry has adopted the concept of scientific and data-driven methodology for quality analysis and improvement. The conventional concepts of sampling inspection, manual visual inspection, and tool-based inspection were not the solution for high-density PCB assembly products. As a result, process engineers and quality control engineers are turning to automated machines for addressing quality and productivity issues.

Fig. 1: AOI is the most cost-effective and reliable inspection tool

Hence, machines for solder paste inspection (SPI), automated optical inspection (AOI), flying probe test, and automatic X-ray inspection have become an important part of PCB assembly manufacturing industry. This article discusses automated optical inspection (AOI) and its applications in the PCB industry.

Why AOI?

Why inspection, tests, and measurements are important during the manufacturing process? The answer is simple—cost reduction. It's crucial to identify issues and defects in the early stages. Any small defect in terms of component placement, lead solder bond, and component functionality can lead to drastic repair cost and field failures. Manufacturers and assemblers prefer high-quality assemblies to ensure no defects.

Automated optical inspection (AOI) is one of the best inspection tools in the SMT environment. Though automatic X-ray inspection is the latest trend, AOI is the most cost-effective and reliable inspection tool. By using one or more HD cameras, this process captures images of assembled PCB surface with the help of a light source. It then compares the captured scanned images with pre-existing template images to detect both catastrophic failures (for example, missing component) and quality defects.

AOI applications

AOI can be applied to both bare board PCBs and PCB assembly lines. Generally, post-reflow AOI is preferred by many process engineers depending on what type of information you are looking for and how fast the inspection needs to be completed.

In the broad sense, AOI must be capable to inspect the presence of an object, recognise barcode information, perform measurement or gauging, package dimension, and lead condition to generate yield rate, distributed defect type, the number of defects, etc.

The following information can be expected from the inspection results:

- Body of machine missing
- Body dimension
- Polarity
- Upside down
- Tombstone
- Billboard
- Pad offset or shift
- Coplanarity